

Muhammad Afsah Mumtaz

AI Engineer | Machine Learning Engineer | LLM & Applied AI Systems

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LinkedIn: linkedin.com/in/muhammad-afsah-mumtaz | GitHub: github.com/NAME0x0 | Portfolio: NAME0x0.github.io

Nationality: Pakistani | Visa: UAE Golden Visa (no sponsorship required) | Open to Dubai / Abu Dhabi (on-site / hybrid) & remote worldwide | Notice period: Immediate

PROFESSIONAL SUMMARY

AI / Machine Learning engineer who builds and ships **large language model (LLM) systems end to end** — fine-tuning, retrieval-augmented generation (RAG), multi-agent orchestration, evaluation, and serving (BSc Hons Information Technology student, Middlesex University Dubai). Fine-tuned a 2B LLM with QLoRA to **82% ARC-Challenge and 92% ARC-Easy — beating Llama 3.2 3B-Instruct — on a single 4 GB GPU**, and shipped a **live 11-agent LLM platform** orchestrating 11 model providers. Deep with Python, PyTorch, Hugging Face, and Ollama; active open-source AI-research contributor. Seeking an **AI Engineer or Machine Learning internship or junior role**. UAE Golden Visa holder — no sponsorship required, immediate start.

TECHNICAL SKILLS

AI / Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Fine-Tuning (LoRA / QLoRA, PEFT), Retrieval-Augmented Generation (RAG), Multi-Agent Systems / Agentic AI, LLM Orchestration, Prompt Engineering, Model Evaluation & Benchmarking, Quantization (GGUF), Inference Optimization, Model Serving, Distillation

AI Frameworks & Tooling: PyTorch, Hugging Face Transformers, PEFT, Ollama, Anthropic Claude, OpenAI, Google Gemini, Groq, DeepSeek, Vector Search (HNSW), SQLite FTS5

Languages: Python, Rust, TypeScript, JavaScript, C++, SQL

Backend & APIs: FastAPI, Node.js, Express, REST API design, API integration, PostgreSQL, Redis, MongoDB, Linux, WSL2

Infra & Tooling: Docker, Git, GitHub Actions, CI/CD, PowerShell, CMake, Release Packaging, Agile, Reproducible Evaluation

PROJECTS

AVA — Local-First LLM Fine-Tuning, Evaluation & Inference Lab github.com/NAME0x0/AVA

Python · PyTorch · Hugging Face · PEFT / QLoRA · RAG · Ollama · GGUF

- Fine-tuned Qwen3.5-2B with a 42 MB QLoRA adapter (0.58% trainable parameters) to **82.0% ARC-Challenge, 92.0% ARC-Easy, and 59.2% MMLU**, outperforming Llama 3.2 3B-Instruct (78.6%) on ARC-Challenge.
- Engineered a local-first ML stack for training, evaluation, distillation, and serving of LLMs on a single 4 GB VRAM laptop GPU.
- Built a reproducible evaluation harness across 17 benchmarks (16,872 tasks); training peaked at 1.81 GB VRAM in 100 minutes, with open-sourced adapters and artifacts.

Pantheon-Trades — Multi-Agent LLM System [Live ↗ pantheon-trades.vercel.app](https://pantheon-trades.vercel.app)

Python · FastAPI · Multi-LLM (Claude, Gemini, OpenAI, Groq, DeepSeek, +6) · Next.js 14 · PostgreSQL · Redis

- Designed and shipped a live system where an 11-agent LLM council deliberates over four structured rounds (opening, challenge, synthesis, blind vote) before any decision.
- Engineered a pluggable multi-LLM backend across 11 providers with self-calibrating agent weights (Platt / isotonic regression) driven by Brier scores.
- Validated with 714 Python and 65 Solidity tests and a 200-market backtest, demonstrating production-grade AI evaluation and reliability.

MALD — Local AI Knowledge System with Cited RAG

Rust · SQLite FTS5 · HNSW vector indexing · RAG · Ollama · Terminal UI

- Built a privacy-first knowledge system as a single Rust binary combining Markdown notes, local AI search, and cited RAG chat with file- and line-level references.
- Implemented hybrid full-text and semantic search with HNSW indexing, backlinks, graph views, and plugin support.

pane — Windows-Native Linux Environment Platform github.com/NAME0x0

Rust · WSL2 · XRDp · PowerShell · Windows packaging automation

- Developed a Windows-native control surface to create, launch, repair, and manage Linux desktop environments — shipped an Arch-first MVP packaged as pane.exe with automated release packaging.

| WORK EXPERIENCE

- Open-Source Contributor — HAGI (Hypercomplex AGI Research)**
github.com/ShmidtS/HAGI | Remote

2026 - Present

 - Run independent experiments on the experimental branch of an open research architecture exploring Clifford-algebra reasoning in small language models — executing and publishing test results across PyTorch, Rust, and Lean4 and exchanging findings with the contributor team.
- Chairman — MDX Computing Society**
Middlesex University Dubai | Dubai, UAE

Sep 2025 - Present

 - Lead the university computing society, directing technical workshops, events, and community initiatives across the IT cohort.
- Committee Chair — BCS Student Chapter**
BCS, The Chartered Institute for IT | Dubai, UAE

Sep 2025 - Present

 - Chair the student chapter committee, planning and delivering student-led technical and professional events.
- Customer Service Representative**
transcosmos inc. | Debrecen, Hungary

Oct 2021 - Jan 2022

 - Resolved inbound customer issues for a multinational outsourcing operation, consistently meeting response-time and quality targets in a high-volume environment.

| EDUCATION

- BSc (Hons) Information Technology**
Middlesex University Dubai | Dubai, UAE

Sep 2024 - Jun 2027 (Expected)

 - Grade 1 (highest grade, UK 20-point scale) in 4 of 6 completed modules; all 6 graded within the first-class band.
 - Relevant coursework: Data Science and Machine Learning, Web Applications and Databases, Programming for Data Communication and Networks, Computer Systems Architecture and Operating Systems, Information Systems Analysis and Design.
- A Levels**
Cambridge International School, Dubai | Dubai, UAE

Completed

| AWARDS & COMPETITIONS

- 2nd Place — Odoo Hackathon (team event), Dubai — Nov 2025
- 2nd Place — Curtin University Dubai Hackathon (team event) — Nov 2025

| CERTIFICATIONS

- IBM AI Engineering Professional Certificate — IBM (Aug 2025)
- AI Strategy and Governance — University of Pennsylvania (Sep 2025)
- Google IT Automation with Python Professional Certificate — Google (Jan 2025)

| ADDITIONAL INFORMATION

Languages: English (Native), Urdu (Native), Arabic (Professional Working), Hungarian (Limited Working)

Work Authorization: UAE Golden Visa holder — no sponsorship required

Availability: Internship, junior, contract, freelance, and full-time. On-site / hybrid in Dubai / Abu Dhabi, UAE, and remote worldwide. Immediate start.